

German Credit Prediction Project | ML Model Results

Executive report

Problem Introduction

Overview

Develop an Machine Learning model for Classification problem. Label credit of German Credit dataset for this project

Problem

Credit Prediction is major for Banks as it helps creating an Automated Decision Support Systems for Credit Progress such as: credit limit, loan approval.

False labeling could lead to Severe Financial Losses, Opportunity Cost & Customer Churn...

Solution

To obtain a model with the highest predictive power, I built three different models to cross-compare results: Logistic Regression, Random Forest and Naive Bayes.

To develop the models, I split a dataset into train (80%) file and test (20%) file.

Details

In the banking and financial services industry, credit provisioning (lending) serves as the primary engine for revenue generation. However, it simultaneously exposes financial institutions to substantial credit risk. The core operational challenge lies in optimizing the trade-off between **portfolio growth** (increasing loan volume) and **risk mitigation** (minimizing capital losses).

The **German Credit Dataset** (originally curated by Prof. Hans Hofmann at the University of Hamburg) provides a benchmark framework containing 1,000 historical loan records. This problem's type is Supervising Learning - specially Binary Classification.

The target variable Risk classifies applicants into two distinct economic regimes:

Good Risk (1): Creditworthy borrowers who exhibit a high probability of repaying both the principal capital and accrued interest within the designated contract duration. These individuals generate stable, predictable cash inflows for the bank.

Bad Risk (0): High-risk borrowers who demonstrate a strong propensity toward delinquency, default, or non-performing status. Approving these applicants results in Non-Performing Loans (NPLs), forcing the bank to absorb direct capital losses and incur massive credit operational costs (collections and legal fees).

In credit scoring, the financial costs associated with the two types of errors are highly asymmetric, making standard evaluation metrics like *Accuracy* fundamentally deceptive.

False Positives (Type I Error in Risk Management): The model inaccurately labels a "bad" credit applicant as "good", resulting in loan approval -> The bank loses the entire cash disbursement, which directly deteriorates the bank's capital adequacy ratio.

False Negatives (Type II Error in Risk Management): The model inaccurately labels a "good" credit applicant as "bad", resulting in loan rejection -> This represents an **Opportunity Cost**. The bank does not lose cash directly from its reserves, but it loses the potential net interest margin that a trustworthy customer would have generated.

German Credit Prediction Project Proposal

Proprietary +
Confidential

Objective

Develop a machine learning model that will help perform predictive Credit Risk on Customers at Bank.

Planning and Analyzing stages

Milestone	Tasks	Outcome/Deliverables
Milestone 1	- Outline project workflow - Import data sources - Explore the dataset	Data Dictionary
Milestone 2	Clean, convert, and format data	Database ready for modeling

Constructing and Executing stages

Milestone 3	- Finalize modeling strategies - Build machine learning models - Test models for accuracy	Machine learning model
Milestone 4	- Finalize results - Share findings in report	- Visualizations - Executive summary - Results

Describe the Dataset

Data Dictionary

Variable Name	Data Type - Variable Type	Values / Ranges	Context & Definition
laufkont	Categorical - Ordinal	1, 2, 3, 4	Status of existing checking account: 1: Little (< 0 DM) 2: Moderate (0 <= ... < 200 DM) 3: Rich (>= 200 DM / salary assignments) 4: No checking account
laufzeit	Integer - Continuous	Months (e.g., 6 - 72)	Duration: Loan repayment period in months.
moral	Categorical - Nominal	0, 1, 2, 3, 4	Credit history: History of compliance with past credit obligations: 0: No credits taken / all credits paid back duly 1: All credits at this bank paid back duly 2: Existing credits paid back duly till now 3: Delay in paying off in the past 4: Critical account / other credits existing (not at this bank)
verw	Categorical - Nominal	0, 1, 2, 3, 4, 5, 6, 8, 9, 10	Purpose: Purpose of the loan requested: 0: New car 1: Used car 2: Furniture/equipment 3: Radio/television 4: Domestic appliances 5: Repairs 6: Education 8: Retraining 9: Business 10: Others
hoehe	Integer - Continuous	DM (Deutsche Mark)	Credit amount: Total credit amount requested by the client.
sparkont	Categorical - Ordinal	1, 2, 3, 4, 5	Savings account/bonds: Level of financial asset accumulation: 1: Little (< 100 DM) 2: Moderate (100 <= ... < 500 DM) 3: Quite rich (500 <= ... < 1000 DM) 4: Rich (>= 1000 DM) 5: Unknown / no savings account
beszeit	Categorical - Ordinal	1, 2, 3, 4, 5	Present employment since: Present employment duration: 1: Unemployed 2: < 1 year 3: 1 <= ... < 4 years 4: 4 <= ... < 7 years 5: >= 7 years
rate	Integer - Ordinal	1, 2, 3, 4	Installment rate: Installment rate in percentage of disposable income.
famges	Categorical - Nominal	1, 2, 3, 4, 5	Personal status and sex: Marital status and biological gender: 1: Male (divorced/separated) 2: Female (divorced/separated/married) 3: Male (single) 4: Male (married/widowed) 5: Female (single)

Data Dictionary

Variable Name	Data Type - Variable Type	Values / Ranges	Context & Definition
buerge	Categorical - Nominal	1, 2, 2003	Other debtors / guarantors: Other parties liable for the loan: 1: None 2: Co-applicant 3: Guarantor
wohnzeit	Integer - Ordinal	1, 2, 3, 4	Present residence since: Duration of living at current residence (Years).
verm	Categorical - Ordinal	1, 2, 3, 4	Property: Most valuable financial or physical asset owned: 1: Real estate 2: Building society savings agreement / life insurance 3: Car or other (not in attribute 1 or 2) 4: Unknown / no property
alter	Integer - Continuous	Years (e.g., 19 - 75)	Age: Age of an individual at the time of loan evaluation.
weatkred	Categorical - Nominal	1, 2, 2003	Other installment plans: Other ongoing payment plans: 1: Bank 2: Stores 3: None
wohn	Categorical - Nominal	1, 2, 2003	Housing: Housing ownership status of the individual:1: Rent 2: Own 3: For free
bishkred	Integer - Ordinal	1, 2, 3, 4	Number of existing credits: Number of existing credits at this bank.
beruf	Categorical - Ordinal	1, 2, 3, 4	Job: Job skill level and residency status of the individual: 1: Unemployed / unskilled - non-resident 2: Unskilled - resident 3: Skilled employee / official 4: Management / self-employed / highly qualified employee
pers	Integer - Binary	1, 2	Number of dependents: Number of people liable to provide maintenance for.
telef	Categorical - Binary	1, 2	Telephone: Telephone registration status under customer's name: 1: None (no telephone registered) 2: Yes (registered under the customer's name)
gastarb	Categorical - Binary	1, 2	Foreign worker: Foreign worker status of the individual: 1: Yes 2: No
kredit	Categorical - Target	1, 2	Credit risk status (Target Variable): 1: Good credit risk (Trustworthy borrower) 2: Bad credit risk (High risk / Potential default)

Exploratory Data Analysis

Objective

To systematically investigate the raw dataset to uncover underlying structures, maximize insights into the data patterns, and detect anomalies before applying any machine learning algorithms.

Key Insights

Details Response

1. Change columns name to English bases on Data Dictionary
2. Splitting Numerical and Categorical features into 2 seperate variables. Plotting the distribution of features to deep dive in.
3. Generating correlation coefficient between each features and plotting correlation matrix in heatmap.

Applying 3 methods of Correlation: Pearson, spearman and Kendall method to figure out what features affect target the most.

4. Group instances based on the classes in outcome variable. Then, calculate the percentage of each value of the OUTCOME variable from total

- There is 21 numerical and 0 categorical features. After visualization, figure out only 3 actual numerical features ('duration', 'credit_amount', 'age') while other are categorical (included target)
- 'checking_account' - 'credit_history' - 'savings_account' have strong positive correlations with target (0.3x - 0.2x - 0.1x) and 'duration' is always the most negative variable (-0.21/-0.21/-0.18)
- There is imbalance between two classes. It shows that around 70% of instances consists 'credit_risk' value as good versus 30% of instances consists bad

Issues/ Problems

1. Columns name is in German
2. Distribution of Numerical and Categorical Features
3. Correlation between features and versus target
4. Check if the data is imbalance

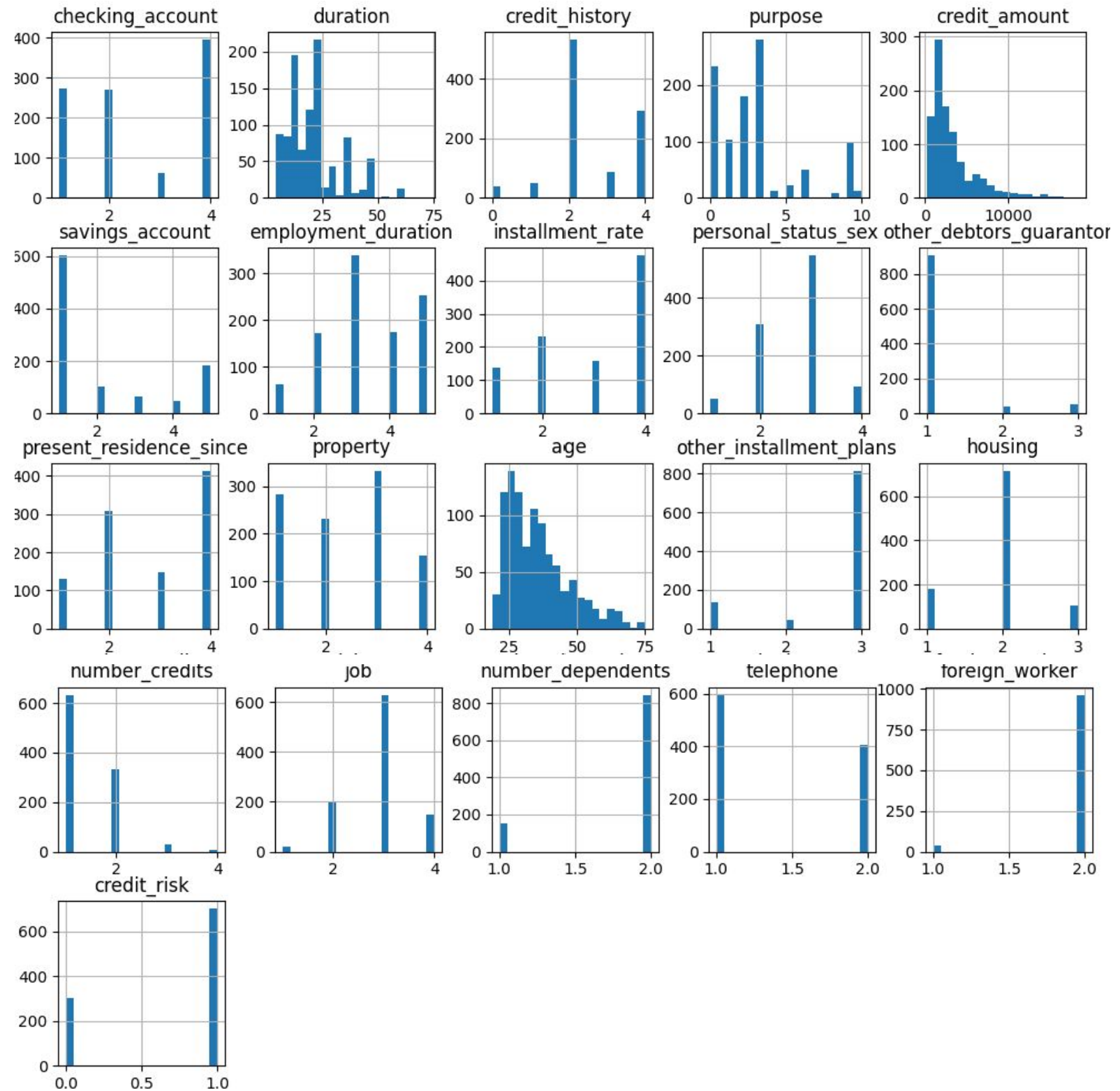
Issue 1

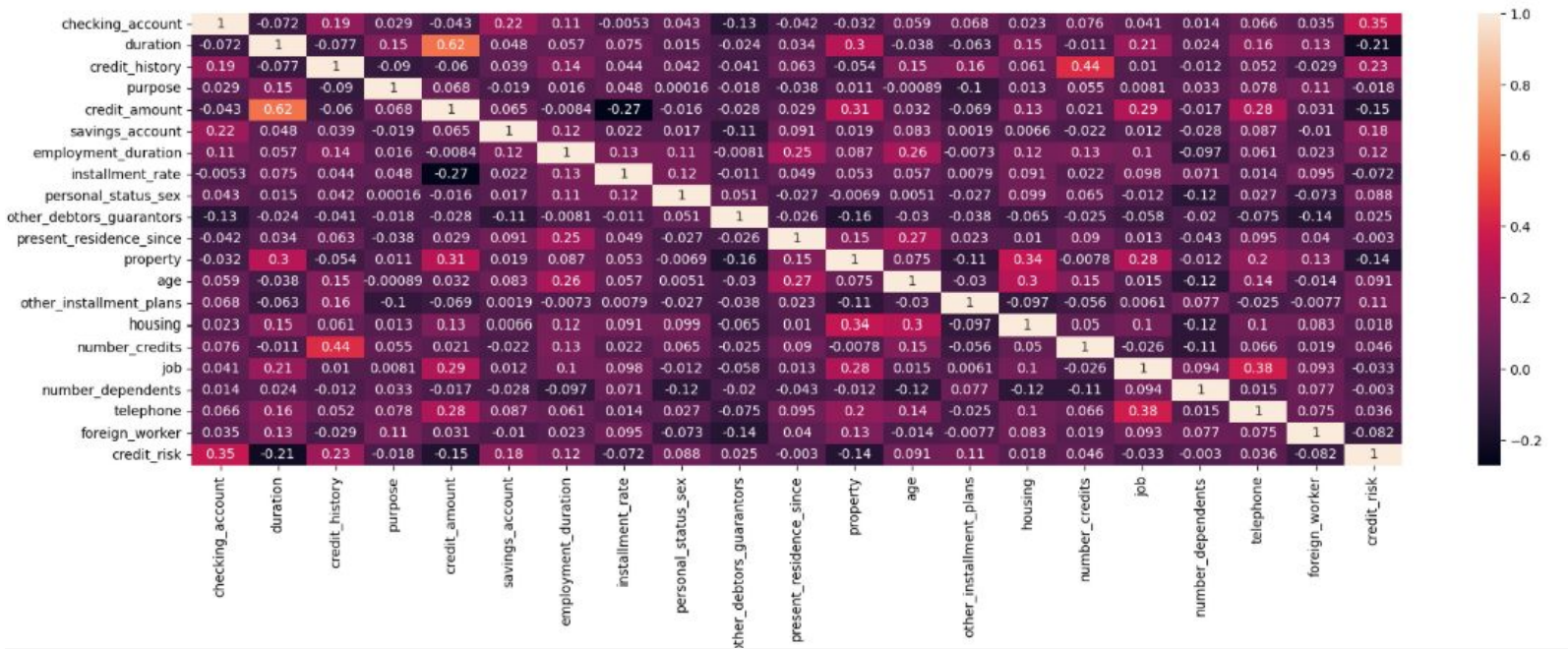
Change Columns name to English

	checking_account	duration	credit_history	purpose	credit_amount	savings_account	employment_duration	installment_rate	personal_status_sex
0	1	18	4	2	1049	1	2	4	2
1	1	9	4	0	2799	1	3	2	3
2	2	12	2	9	841	2	4	2	2
3	1	12	4	0	2122	1	3	3	3
4	1	12	4	0	2171	1	3	4	3

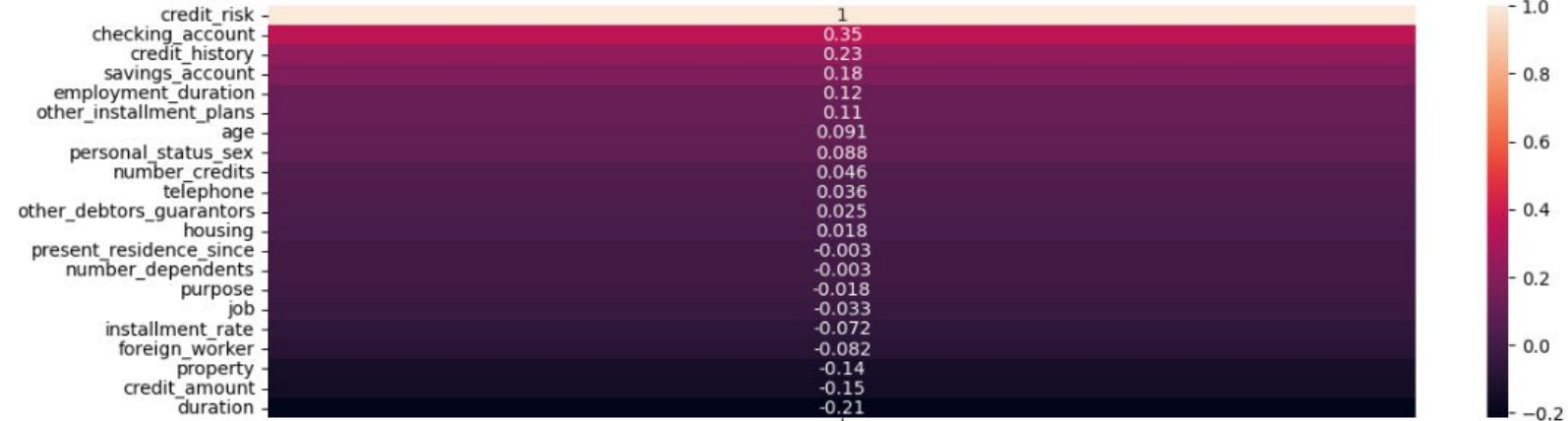
Issue 2

Distribution of Numerical and Categorical Features

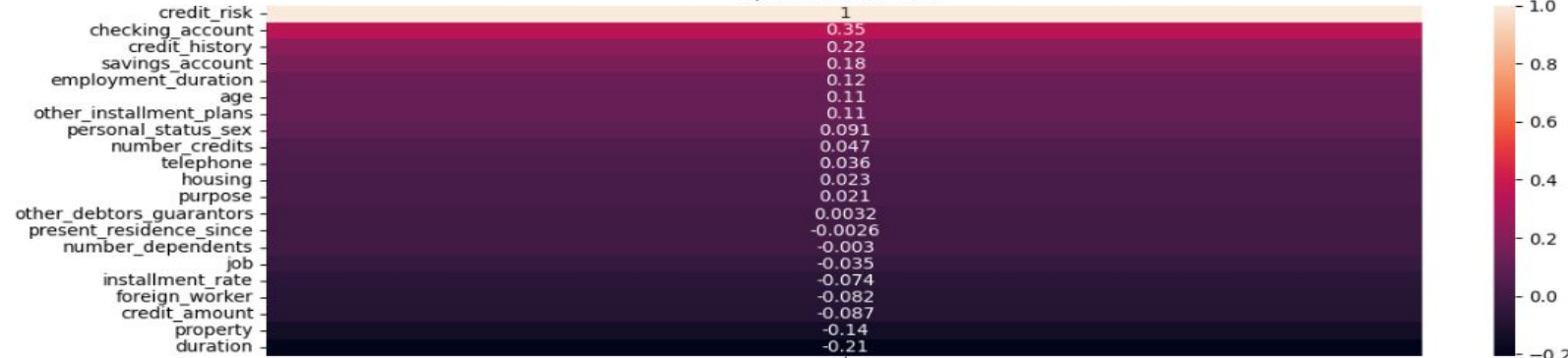




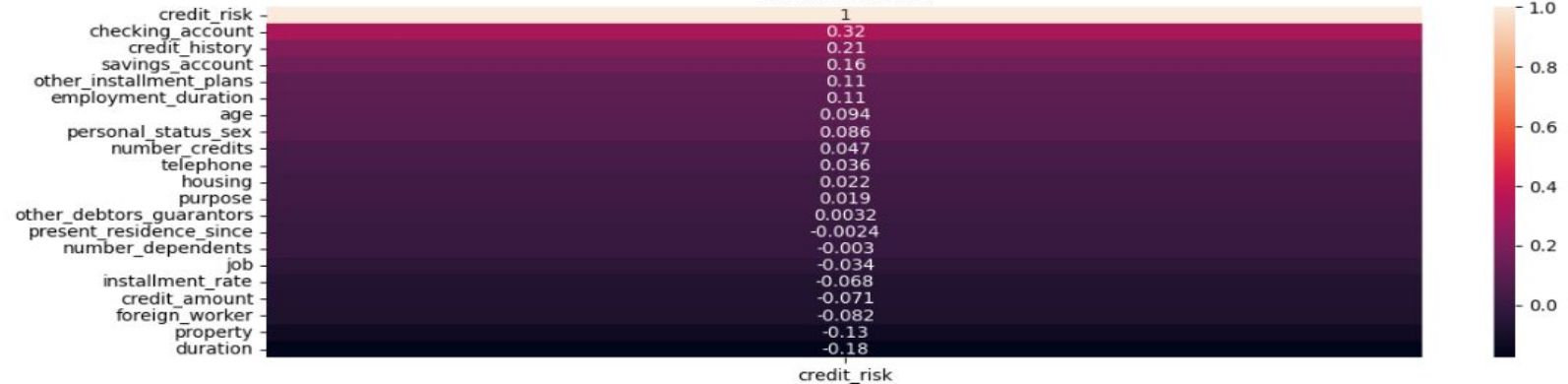
Pearson method



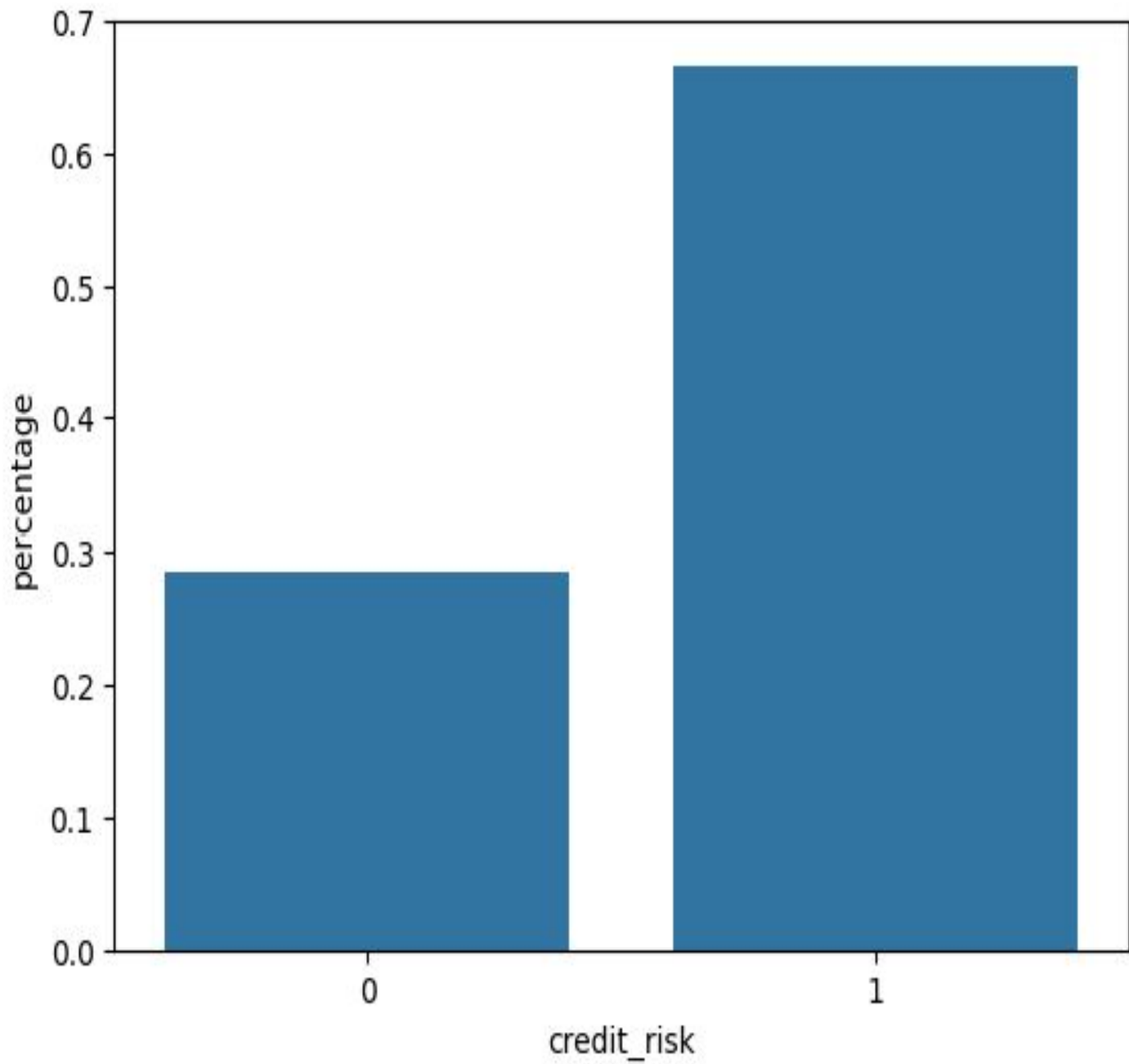
spearman method



Kendall method



	credit_risk	count	percentage
0	0	300	0.286
1	1	700	0.667



Preprocessing

Objective

Dataset ready for modeling

Key Insights

Details Response

1. Check all aspects of Data Quality. Synchronize Yes - No information in 3 columns: telephone, foreign_worker and credit_risk
2. Use One-Hot Encoding and Ordinal Encoding and Split features into Ordinal features and Nomials features based on Data Dictionary
3. Split Train (80%) and Test (20%) thanks to available function in scikit_learn
4. Scaling Numerical features on Train by using function fit_transform from scikit-learn and use only transform on Test set
5. Applying Handling Imbalance SMOTE on Train only then fit and resample on train only after scale

- There is a confusion in value 'Yes - No' in telephone, foreign_worker and credit_risk. Other are perfect

Issues/ Problems

1. Data Quality includes Completeness, Accuracy, Validity & Consistency, Timeless, Uniqueness
2. Categorical Features Encoding
3. Train/Test Data Splitting
4. Feature Scaling for Numerical Data
5. Handling Data Imbalance

Preprocessing

Objective

Dataset ready for modeling

Value

Details Response

1. Check all aspects of Data Quality. Synchronize Yes - No information in 3 columns: telephone, foreign_worker and credit_risk
2. Use One-Hot Encoding and Ordinal Encoding and Split features into Ordinal features and Nomials features based on Data Dictionary
3. Split Train (80%) and Test (20%) thanks to available function in scikit_learn
4. Scaling Numerical features on Train by using function fit_transform from scikit-learn and use only transform on Test set
5. Applying Handling Imbalance SMOTE on Train only then fit and resample on train only after scale

- There is a confusion in value 'Yes - No' in telephone, foreign_worker and credit_risk. Other are perfect.
- Due to the imbalance of dataset, after splitting the Train set is also imbalance. After handling, the Train set is good now

Label distribution on Train befor SMOTE:
credit_risk

1 560

0 240

Name: count, dtype: int64

Label distribution on Train befor SMOTE:
credit_risk

0 560

1 560

Name: count, dtype: int64

Issues/ Problems

1. Data Quality includes Completeness, Accuracy, Validity & Consistency, Timeless, Uniqueness
2. Categorical Features Encoding
3. Train/Test Data Splitting
4. Feature Scaling for Numerical Data
5. Handling Data Imbalance

Models Building

Reason why choosing

Models usage

Three different models to cross-compare results: Logistic Regression, Random Forest and Naive Bayes.

Logistic Regression

Random Forest

German Credit dataset contains mostly numerically coded categorical variables and skewed variables such as Credit Amount. Random Forest is extremely capable of handling this mixed data without being affected by the scale or distribution of the variables

By combining hundreds of decision trees through the Bagging mechanism, this algorithm effectively controls overfitting on a small dataset of 1,000 rows, while automatically capturing complex non-linear interactions between customer features

Random Forest is a non-parametric algorithm. It is completely unaffected by the strong correlation between financial variables that we discovered in the EDA step, ensuring the model operates stably and without noise.

- This is a classic binary classification algorithm, serving as an ideal baseline model for evaluating the performance of more complex models later on
- The German Credit dataset is relatively small (1,000 rows), so simple linear models like Logistic Regression help control overfitting risk very well
- Finally, in the credit scoring problem of the banking industry, interpretability is essential. Logistic Regression allows to understand the weight and impact of each variable (such as account balance or loan term) on the probability of customer bad debt

Naive Bayes

I needed to experiment with different mathematical approaches. After selecting the 2 former, Naive Bayes proved to be the best representative of the generative probabilistic model

The dataset is quite small. The simple mathematical nature of Naive Bayes makes this model immune to overfitting. Including this model helps me test whether the assumption of data independence is appropriate for this credit risk problem.

Model Evaluation

Objective

Assess the model Performance
Logistic Regression, Random Forest and Naive Bayes

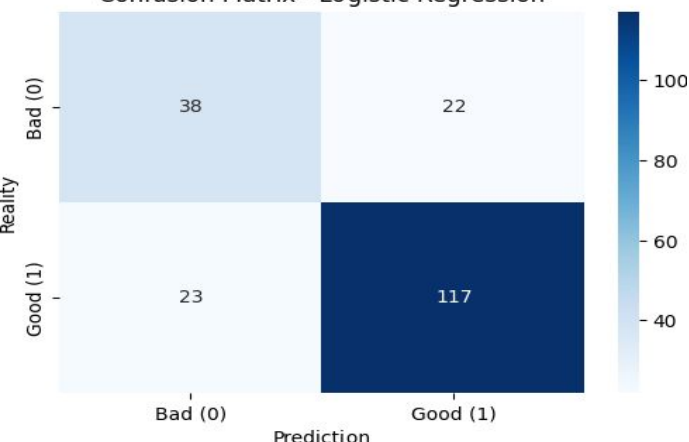
Logistic Regression

Metrics

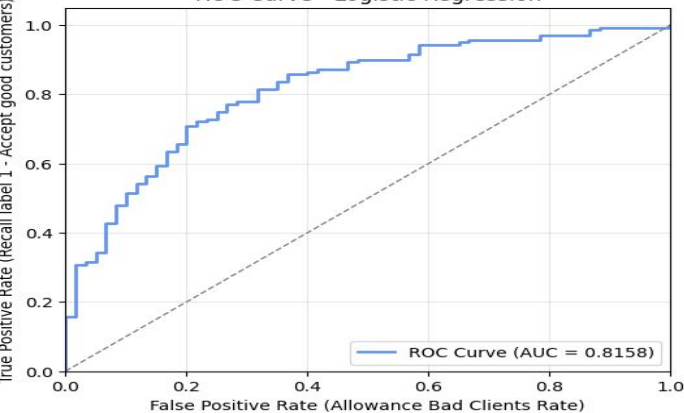
CLASSIFICATION REPORT

	precision	recall	f1-score	support
0	0.62	0.63	0.63	60
1	0.84	0.84	0.84	140
accuracy			0.78	200
macro avg	0.73	0.73	0.73	200
weighted avg	0.78	0.78	0.78	200

Confusion Matrix - Logistic Regression



ROC Curve - Logistic Regression



The model achieved a solid ROC-AUC score of 0.8158, paired with an overall accuracy of 78%. The macro and weighted averages stood at 73% and 78%, respectively, indicating an acceptable predictive boundary with no severe signs of overfitting or underfitting.

In terms of class-specific metrics, In identifying creditworthy applicants (good class), the model yielded a high and balanced Recall of 84% and Precision of 84%. This balanced performance proves that the model's good customer identification is highly sensitive and accurate.

Conversely, the metrics for the high-risk applicants (bad class) fell into an upper-intermediate range, with a Recall of 63% and a Precision of 62%. From a risk management perspective, this implies that 37% of high-risk applicants (representing 22 individuals) managed to slip through the system's filter into the loan approval pool. Concurrently, the model mistakenly flagged a portion of creditworthy applicants as high-risk, resulting in the erroneous rejection of 23 good and potential clients

Model Evaluation

Objective

Assess the model Performance
Logistic Regression, Random Forest and Naive Bayes

Random Forest

Metrics

The model achieved a solid ROC-AUC score of 0.8315, paired with an overall accuracy of 79%. The macro and weighted averages stood at 74% and 79%, respectively, indicating an acceptable predictive boundary with no severe signs of overfitting or underfitting.

In terms of class-specific metrics, In identifying creditworthy applicants (good class), the model yielded a high rate as Recall of 81% and Precision of 88%. This balanced performance proves that the model's good customer identification is highly sensitive and accurate.

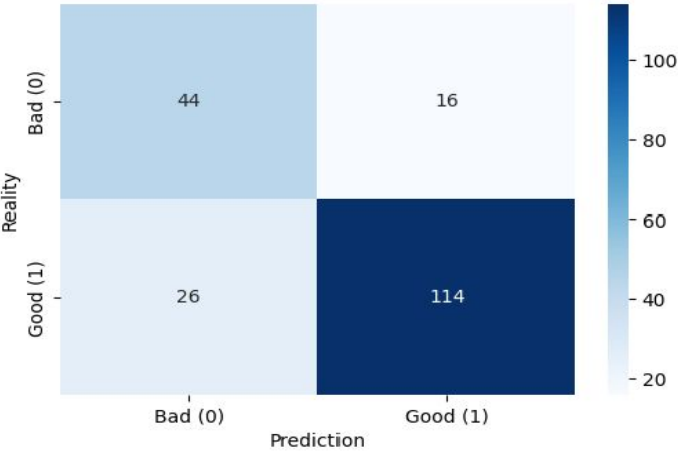
Conversely, the metrics for the high-risk applicants (bad class) fell into an upper-intermediate range, with a Recall of 73% and a Precision of 63%. From a risk management perspective, this implies that 27% of high-risk applicants (representing 16 individuals) managed to slip through the system's filter into the loan approval pool. Concurrently, the model mistakenly flagged a portion of creditworthy applicants as high-risk, resulting in the erroneous rejection of 26 good and potential clients

CLASSIFICATION REPORT - RANDOM FOREST

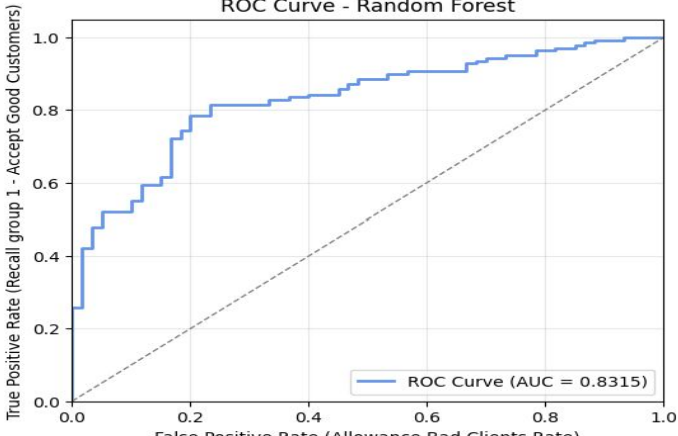
	precision	recall	f1-score	support
0	0.63	0.73	0.68	60
1	0.88	0.81	0.84	140
accuracy			0.79	200
macro avg	0.75	0.77	0.76	200
weighted avg	0.80	0.79	0.79	200

ROC-AUC Score: 0.8315

Confusion Matrix - Random Forest



ROC Curve - Random Forest



Model Evaluation

Objective

Assess the model Performance
Logistic Regression, Random Forest and Naive Bayes

Naïve Bayes

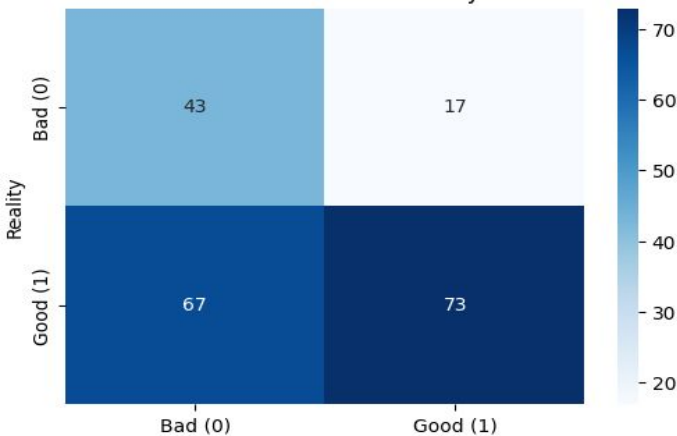
Metrics

CLASSIFICATION REPORT - NAIVE BAYES

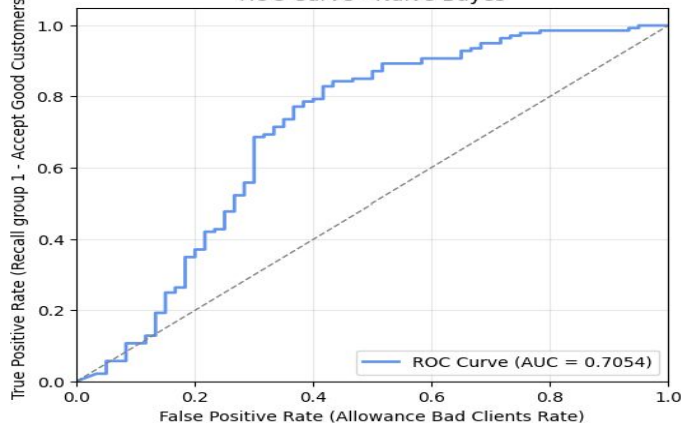
	precision	recall	f1-score	support
0	0.39	0.72	0.51	60
1	0.81	0.52	0.63	140
accuracy			0.58	200
macro avg	0.60	0.62	0.57	200
weighted avg	0.69	0.58	0.60	200

ROC-AUC Score: 0.7054

Confusion Matrix - Naive Bayes



ROC Curve - Naive Bayes



The model achieved a solid ROC-AUC score of 0.7054, paired with an overall accuracy of 58%. The macro and weighted averages stood at 57% and 60%, unusual, model seems to be underfitting.

In terms of class-specific metrics, In identifying creditworthy applicants (good class), the model yielded an unbalance rate as Recall of 52% and Precision of 81%. This performance proves that the model good customer identification is unsensitive so it accept a large of bad customer (17 actual bad customers to slip through the net)

Moreover, the metrics for the high-risk applicants (bad class) are much more uneven, with a Recall of 72% and a Precision of 39%.

The recall metric is good but the model tend to be overreact so it accused a large number of good clients to bad risk (67 good and potential clients were mistakenly denied)

Result Comparison

Performance Comparison Table

	Model	ROC-AUC	Accuracy	Precision (Bad-0)	Recall (Bad-0)	Precision (Good-1)	Recall (Good-1)	Macro Avg F1
0	Logistic Regression	0.816	0.780	0.620	0.630	0.840	0.840	0.730
1	Random Forest	0.832	0.790	0.630	0.730	0.880	0.810	0.760
2	Naive Bayes	0.705	0.580	0.390	0.720	0.810	0.520	0.570

Assessment

Overview

- Random Forest is the most optimal models in these three
- Logistic Regression is the second one
- Naive Bayes is the least due to underfitting

Details

Model	Advantages	Disadvantages	Description
Logistic Regression	Accuracy is quite high (0.78) ROC-AUC is good (0.816)	Low Recall (Bad) (0.62)	Easy to explain to regulators and the Risk department For banks, this is a relatively high rate of omissions.
Random Forest	Highest ROC-AUC (0.832) Highest Accuracy (0.79) Highest Macro F1 (0.76) Highest Recall(Bad) (0.73)		This model strikes the best balance between: Risk detection capability Overall accuracy Ability to differentiate between two classes
Naive Bayes	Recall (Bad) = 0.72 is close to Random Forest	Precision (Bad) = 0.39 (very low) Accuracy = 0.58 ROC - AUC = 0.705	Turning away many good customers Loss of revenue Poor customer experience

Conclusion & Recommendations

Conclusion

For the German Credit Risk problem, the bank's goal is not just to achieve the highest Accuracy, but to minimize the risk of granting credit to bad customers.

Therefore, in credit risk management, the most important indicator is usually: Recall (Bad) because it measures the ability to detect risky customers.

Random Forest is the most suitable model for credit risk management objectives because it achieves the best balance between the ability to detect risky customers (Recall Bad = 0.73) and overall performance (ROC-AUC = 0.832, Accuracy = 0.79). Logistic Regression is a good alternative when explanatory power is prioritized. Naive Bayes is not suitable for implementation due to its excessively high false alarm rate.

Recommendations

1. Deploy Random Forest as the Primary Credit Risk Model
2. Optimize the Decision Threshold
3. Use Risk-Based Lending Rather Than Binary Approval
4. Keep Logistic Regression as a Benchmark Model
5. Incorporate Additional Customer Information
6. Regularly Monitor and Retrain the Model
7. Future Work: Applying advanced Models for the Dataset (Gradient Boosting, XGBoost, LightGBM, CatBoost, etc)